GEK-85708

INSTRUCTIONS

AFTROL* II

Adjustable Speed Drive Controller

Supplement to GEK-85702 for 20HP Square Wave Inverter

*TRADEMARK OF GENERAL ELECTRIC COMPANY, U.S.A.

These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company.

GENERAL ELECTRIC

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INTRODUCTION

This book applies to the following AFtrol II* drive controllers:

HP	INPUT VOLTAGE	ENCLOSURE	MODEL NUMBERS
20	230V—3 Ph	Chassis Mtd	331X659AAG01

This manual is a supplement of the basic AFtrol II 10 to 15 HP drive Instruction Book GEK-85702, and covers the special ratings shown in the table above.

For specific information about drive operation detailed description, start-up and adjustments, maintenance and troubleshooting procedures refer to GEK-85702.

SAFETY RECOMMENDATIONS

Qualified electrical and electronics personnel should install and maintain this equipment. They should read the complete instructions prior to applying power or troubleshooting the equipment. They should heed all WARNING and CAUTION notes or labels listed in this Manual or posted on the equipment. Definitions of label terms and colors are as follows:

WARNING

DENOTES OPERATING PROCEDURES AND PRACTICES THAT MAY RESULT IN PERSONAL INJURY OR LOSS OF LIFE IF NOT CORRECTLY FOLLOWED.

COLOR: BLACK OR WHITE LETTERING ON RED FIELD.

CAUTION

DENOTES OPERATING PROCEDURES AND PRACTICES THAT, IF NOT STRICTLY OBSERVED, MAY RESULT IN DAMAGE TO OR DESTRUCTION OF THE EQUIPMENT.

COLOR: BLACK LETTERING ON AMBER FIELD.

NOTE

DENOTES AN OPERATING PROCEDURE OR CONDITION WHICH SHOULD BE HIGHLIGHTED.

COLOR: BLACK LETTERING ON WHITE FIELD.

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RECEIVING HANDLING AND STORAGE

WARNING

IMPROPER LIFTING PRACTICES CAN CAUSE SERIOUS OR FATAL INJURY. LIFT ONLY WITH ADEQUATE EQUIPMENT AND TRAINED PERSONNEL.

Refer to GEK-85702 for information on receiving, handling, and storage for this equipment.

INSTALLATION AND APPLICATION GUIDELINES

LOCATION AND MOUNTING

WARNING

EXPLOSION OR FIRES MAY RESULT FROM MOUNTING THE CONTROLLER IN HAZARDOUS AREAS SUCH AS LOCATIONS WHERE INFLAMMABLE OR COMBUSTIBLE VAPORS OR DUSTS ARE PRESENT. AFTROL II CONTROLLERS MUST BE INSTALLED AWAY FROM HAZARDOUS AREAS EVEN IF USED WITH AC MOTORS SUITABLE FOR USE IN SUCH LOCATIONS.

WARNING

IMPROPER LIFTING PRACTICES CAN CAUSE SERIOUS OR FATAL INJURY, LIFT ONLY WITH ADEQUATE EQUIPMENT AND TRAINED PERSONNEL.

Refer to the outline drawing shown in the figure listed below:

DRIVE MODEL NO.	HP	FIG. NO.	OUTLINE DWG. NO.
331X659AAG01	20	1	36B605405FF

ELECTRICAL CONNECTIONS

WARNING

DANGER OF ELECTRICAL SHOCK OR BURN, DO NOT SERVICE THE EQUIPMENT WHEN THE POWER IS APPLIED.

CAUTION

CARE SHOULD BE TAKEN TO SEE THAT ALL INTERCONNECTION WIRING IS SIZED AND INSTALLED IN ACCORDANCE WITH THE LATEST EDITION OF THE NATIONAL ELECTRICAL CODE (NEC) AND BE CONSISTENT WITH ALL OTHER APPLICABLE LOCAL CODES.

CAUTION

MEGGERING CAN DAMAGE ELECTRONIC COMPONENTS. DO NOT MEGGER OR HI POT WITHOUT CONSULTING THE SPEED VARIATOR PRODUCTS OPERATION, GENERAL ELECTRIC COMPANY.

WARNING

DANGER OF ELECTRICAL SHOCK OR BURN. ALL MOTOR BASES AND CONTROLLER HOUSING SHOULD BE CONNECTED TO THE INSTALLATION EARTH GROUNDING SYSTEM. AN EXTERNAL COPPER CONNECTION IS LOCATED AT THE BOTTOM LEFT SIDE OF THE CONTROLLER AND IS SUITABLE FOR CONNECTING A GROUND WIRE BY MEANS OF A COPPER LUG.

CAUTION

BE SURE TO PROTECT THE INTERNAL PRINTED CIRCUIT CARDS AND OTHER COMPONENTS FROM METAL PARTICLES WHEN CUTTING OR DRILLING ENTRANCES FOR INTERCONNECTION WIRING AND CABLES.

IT IS RECOMMENDED THAT A GREENLEE PUNCH BE USED FOR CUTTING CABLE OPENINGS IN THE ENCLOSURE.

NOTE

IT IS RECOMMENDED THAT THE CONTROL COMMON CIRCUIT BE GROUNDED AT ONLY ONE POINT WHICH IS 2TB22. IF THE SPEED REFERENCE IS SUPPLIED BY A PROCESS INSTRUMENT, THERE SHOULD BE NO MORE THAN 45 VOLTS BETWEEN ITS SIGNAL OUTPUT AND GROUND. (REFER TO FIGURE 16 IN GEK-85702)

CAUTION

DO NOT USE POWER FACTOR IMPROVEMENT CAPACITORS WITH THIS EQUIPMENT WITHOUT CONSULTING SPEED VARIATOR PRODUCTS OPERATION, GENERAL ELECTRIC COMPANY. DAMAGE MAY RESULT FROM HIGH VOLTAGES GENERATED WHEN CAPACITORS ARE SWITCHED.

NOTE

THE AFTROL II DRIVE CONTROLLER **WILL NOT** OPERATE UNLESS THE AC MOTOR IS ELECTRICALLY CONNECTED. DISCONNECTION OF ANY ONE MOTOR PHASE WILL PRODUCE A FAULT SHUTDOWN.

WARNING

ELECTRIC SHOCK CAN CAUSE PERSONAL INJURY OR LOSS OF LIFE WHETHER THE AC SUPPLY IS GROUNDED OR NOT. HIGH VOLTAGES TO GROUND WILL BE PRESENT IN BOTH THE POWER AND CONTROL CIRCUITS AFTER AC SUPPLY POWER IS DISCONNECTED. WAIT AT LEAST ONE MINUTE FOR THE DC LINK CAPACITOR TO DISCHARGE BEFORE TOUCHING ANY PART OF THE CONTROLLER CIRCUITRY.

WARNING: HIGH VOLTAGE

ELECTRIC SHOCK CAN CAUSE PERSONAL INJURY OR LOSS OF LIFE WHETHER THE AC SUPPLY IS GROUNDED OR NOT. HIGH VOLTAGE TO GROUND WILL BE PRESENT AT MANY POINTS. WHEN INSTRUMENTS SUCH AS OSCILLOSCOPES ARE USED TO WORK ON LIVE EQUIPMENT, GREAT CAUTION MUST BE USED WHEN ONE OF THE INSTRUMENT LEADS IS CONNECTED TO THE CASE OR OTHER METAL PARTS OF THE INSTRUMENT. THIS LEAD SHOULD NOT BE CONNECTED TO AN UNGROUNDED PART OF THE SYSTEM UNLESS THE INSTRUMENT IS INSULATED FROM GROUND AND ITS METAL PARTS TREATED AS LIVE EQUIPMENT. USE OF AN INSTRUMENT HAVING BOTH LEADS ISOLATED FROM THE CASE PERMIT GROUNDING OF THE CASE EVEN WHEN THE MEASUREMENTS MUST BE MADE BETWEEN LIVE PARTS.

CAUTION

WHEN TAKING READINGS WHILE THE CONTROLLER IS OPERATING, TAKE CARE NOT TO SHORT BETWEEN CIRCUITS WITH THE METER PROBES. USE INSULATING SLEEVES BETWEEN PRINTED CIRCUIT CARD CONNECTION POINTS AS MEANS TO PREVENT SHORTING.

NOTE

ALWAYS READ THE COMPLETE INSTRUCTIONS PRIOR TO APPLYING POWER OR TROUBLE-SHOOTING THE EQUIPMENT.

READ AND HEED ALL WARNING, CAUTION AND NOTE LABELS POSTED ON THE EQUIPMENT.

Refer to GEK-85702 and the drive input ratings below for input power wiring:

DRIVE MODEL NO.	HP	NO. OF INPUT PHASES	MAX. INPUT AMPS (RMS)
331X659AA001	20	3	59

Use Figure 15A in GEK-85702 for the input power connections.

Refer to GEK-85702 for motor and control wiring.

The System diagrams and Power module elementary diagrams shown in Figures 8A, 9, 9A and 11, in GEK-85702 apply to the drives covered by this instruction book.

OPTION KITS

The option kits listed in GEK-85702 for the 15 HP chassis mounted AFtrol II drive are applicable to the drives covered in this instruction book with the exception of the dynamic braking kit.

A standard dynamic braking kit is not available for 20 HP controllers. Consult the factory for 20 HP dynamic braking applications.

BRIEF DESCRIPTION

MODEL NO. 331X659AAG01
This drive is similar to the standard 15 HP chassis mounted AFtrol II drive controller (6VGAC2015C1) described in GEK-85702, with upgraded fuses and power module to accomodate the special 20 HP rating.

The fuse holders have been changed to 1/4 inch stud type terminal blocks to accomodate the larger fuses. (See the Spare and Renewal Parts section of this instruction book for the fuse catalog number)

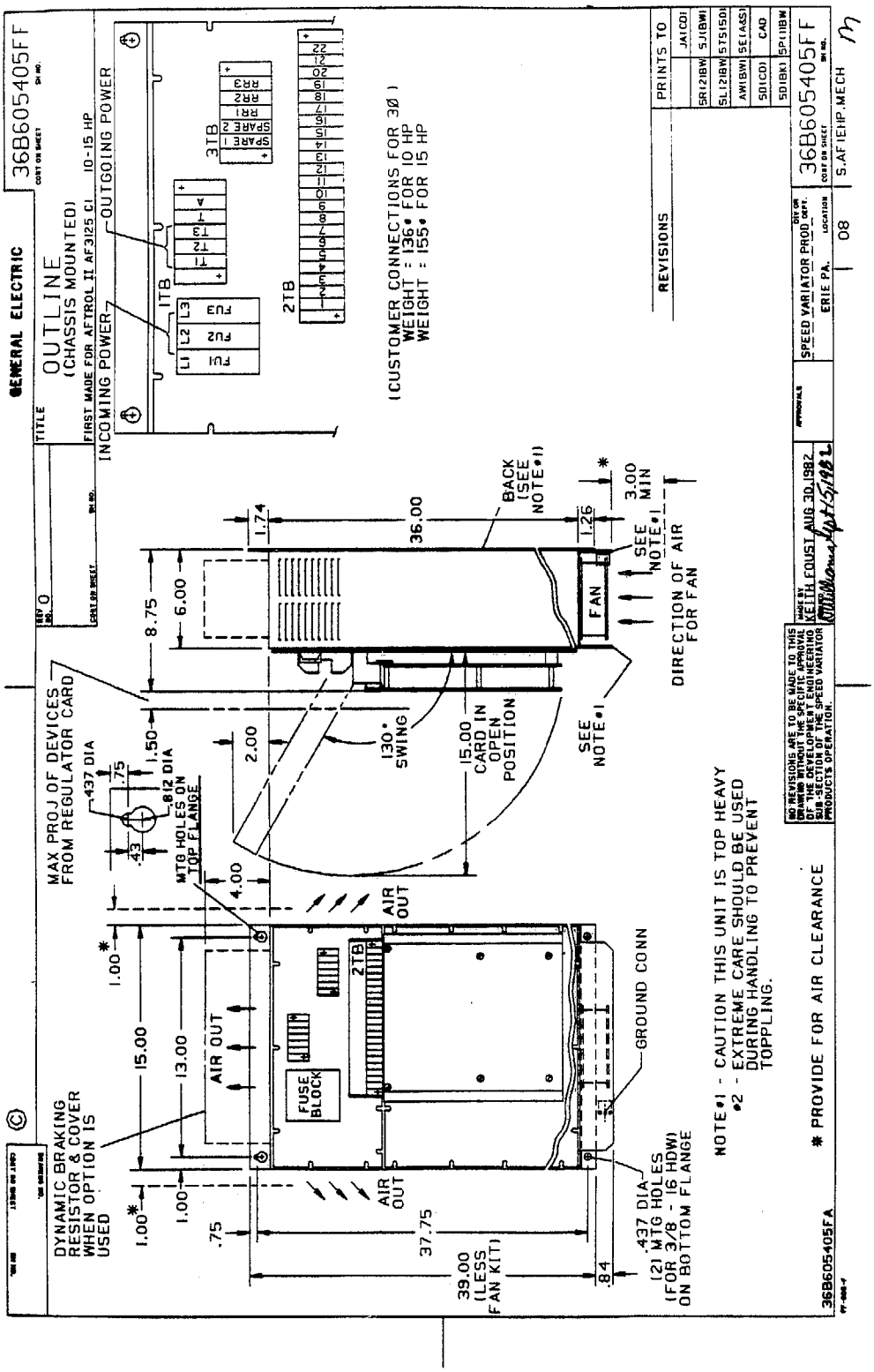


FIGURE 1
20 HP AFtrol Drive Outline Drawing — Chassis Mounted
(36B605405FF)

STANDARD SPECIFICATIONS

RATINGS

MODEL NUMBER 331X659AAG01

Horsepower — 20 HP

Power Supply
Voltage — Three phase, 230V + 10%, -5%
Frequency — 50 or 60 Hertz ±2%

Power Output — 3 phase, 0 to 230 volts Base Frequency at 230V can be adjusted from 50 to 200 Hz (12 ranges)

MODEL NO.	HP	FULL LOAD AMPS INPUT	AMPS RMS OUTPUT
331X659AAG01	20	59	56

Overload Capability
Service Factor — 1.0
1 Minute Overload — 63 Amps RMS Maximum
Breakway Torque — 125%
(NEMA B Induction Motor)

Speed Range — 10 to 1 within frequency range selected i.e. 6 to 60 Hz, 12 to 120 Hz

Service Conditions — Ambient temperature — 0 to 40°C
Altitude — to 3300 feet
— (0 to 1005 meters)

Functions of Basic Controller —
Drive start and stop
Speed control (by potentiometer)
Unidirectional operation
Adjustable linear timed acceleration and deceleration (2 to 20 sec. individual adjustments)

Jumper selectable fast linear time rate (0.4 to 4.0 sec).
Up to 2:1 constant HP range with adjustable crossover from constant torque range.

Speed Regulation — 3% with induction motor
(Due to motor slip)
— 0% with sync. reluctance motor
service deviation — 1%

Adjustments — ACCEL linear time
DECEL linear time
Volts per hertz
Minimimum frequency
Maximum speed
Minimum speed
Voltage Boost
Motor Overcurrent trip level
Crossover to constant HP range

Protective Features —
AC line, fuses
AC line, surge protection
AC line, undervoltage shutdown and loss of phase
Phase insensitive
DC link overvoltage trip
Control undervoltage trip
Motor slowdown limit
(Maintains inverter—motor synchronism during stopping)
Motor stabilization circuit
Motor overtemperature (when used — see 2TB terminal board, points 1 and 2)
Isolation between power and operator's control circuits

Indication:
RUN
FAULT trip
RPS regulator power supply
CPS converter power supply
Converter SCR operation
Inverter Transistor operation
IPS inverter power supply
IF inverter fault trip
UV AC supply undervoltage
OC overcurrent trip

Operator's Station
Start—Stop pushbuttons (control voltage 24 volts DC)
Speed adjust potentiometer
NEMA 1 Enclosure (Refer to Figure 4 in GEK-85702)

SPARE AND RENEWAL PARTS

AFtrol II Drive Controller — SPECIAL RATINGS
Model No. & HP
331X659AAG01 — 20 HP

Catalog Number	Description
193X457AAG01	Snubber Card
193X458AAG01	Amplifier Card
193X460ABG02	Inverter Control Card
193X461ADG01	Regulator Card
193X462AAG01	System Follower Card
193X463AAG01	Speed Readout Card
193X459AAG02	Dynamic Braking Card

SUB—ASSEMBLIES AND COMPONENTS (not HP related)

Catalog Number	Description
36A353243BDG01	Control Transformer with connected wire harness
104X156CA014	Control Transformer only
36B605400ABG03	Ventilating Fan with connected wire harness
104X215CA001	Ventilating Fan only

SPARE AND RENEWAL PARTS (HP RELATED)

AFtrol II Drive Controller - SPECIAL RATINGS

		QTY PER DRIVE
		20 HP
Part Name	Catalog Number	331X659AAG01
AC Power Fuses (100 amp)	104X190AG009	3
Power Module	36D877451FAG02	1
SCR—Diode Module	104X125DC008	3
Diode Module	104X125DC009	1
Power Transistor	104X170CB009	6
Filter Choke	104X220AA013	4
Filter Capacitors	104X122AA355	4

HOTLINE TELEPHONE NUMBER

The Contract Warranty for AFtrol II drives is stated in the General Electric Apparatus Handbook, Section 105, page 71.

The purpose of the following is to provide specific instructions to the AFtrol II drive user regarding warranty administration and how to obtain assistance on out-of-warranty failures.

AFtrol II DRIVE CONTROLLERS
The warranty covers all major parts of the controller such as main printed circuit boards, SCR modules, etc. but does not provide for replacement of fuses or the complete controller.

1. In the event of failure or misapplication during "in warranty" refer to the instruction book to identify the defective part or subassembly.

2. When the defective part has been identified (or for assistance in identification) call:

GENERAL ELECTRIC COMPANY
ERIE, PENNSYLVANIA 16531
(814)—455—0521

(24 hour phone service)

Before calling, list catalog numbers of the controller, motor, operator's station and any modification kits for ready reference.

AFtrol II DRIVE MOTORS
AFtrol II drive motor repairs are generally handled by General Electric Small Motor Service Centers of Apparatus Service Shops. For specific instructions on your motor, call General Electric at the number listed above and furnish complete nameplate data.

GENERAL ELECTRIC COMPANY
SPEED VARIATOR PRODUCTS OPERATION
ERIE, PENNSYLVANIA 16531

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GEK-85708 (8/84) 500 (P)